THE HUMAN EYE & COLOURFUL WORLD

1. The Human Eye

(i) Lens form an image on light sensitive screen called retina

(ii) light enters thru a thin membrane → cornea forms transparent bulge on the front surface of eye ball

most refraction occurs at outer surface of cornea

(iii) diameter = 2.3 cm

(iv) Iris

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(v) retina: light sensitive cells tnt
↓
- generate electrical signals upon illumination
- sent to brain via optic nerves

2. POWER OF ACCOMMODATION

↳ to see distant objects -
ciliary muscles relaxed
lens is thin
focal length ↑↑

↳ to see nearby objects -
ciliary muscles contract
lens → thick
focal length ↓

3. least distance of distinct vision :- Min. distance, at which objects can be seen most distinctly w/o strain
↳ also called near pt. of eye

4. Cataract : crystalline lens becomes milky & cloudy
↓
partial/ complete loss of vision

5. Defects of Vision

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(i) Myopia (near-sightedness)

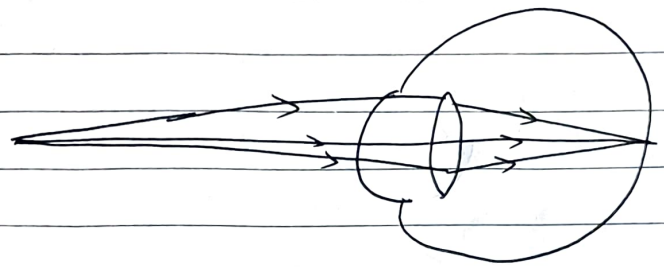
↳ cannot see distinct objects

REASON

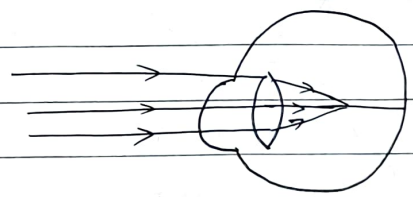
↳ (i) excessive curvature of eye lens

(ii) elongation of eyeball

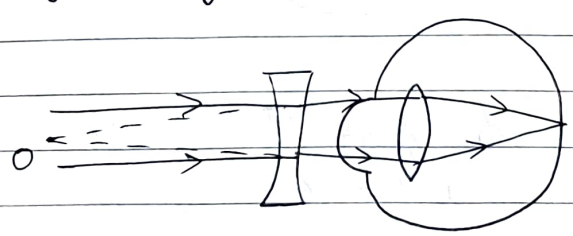
CORRECT: concave lens



(a) Far pt. of myopic eye



(b) Myopic eye



(c) Correction

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(b) Hypermetropia (far-sightedness)

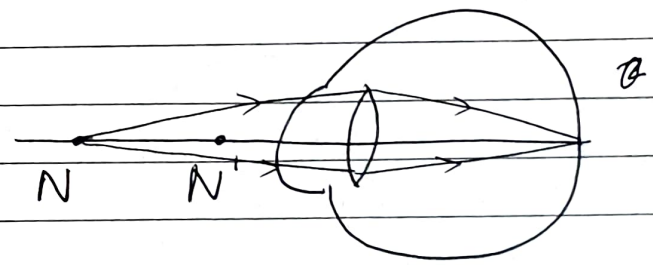
↳ cannot see nearby object

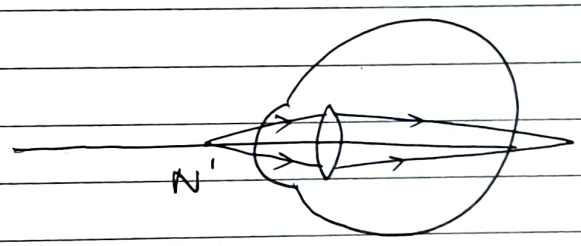
↳ near pt. more than 25cm

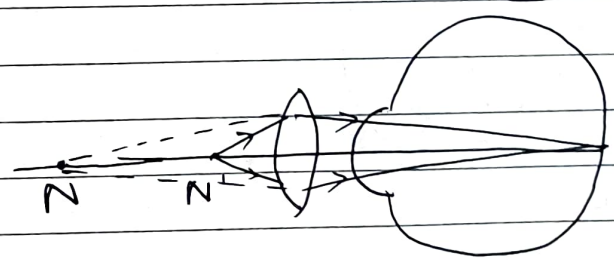
CAUSE (i) eyeball has become small

(ii) focal length of lens ↑

CORRECT:- convex lens







N = Near pt. of hypermetropic eye

N' = Near pt. of normal eye

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(c) Presbyopia

↳ Power of Accomodation ↓

CAUSE (i) weakening of ciliary muscles

(ii) diminishing flexibility of eye lens

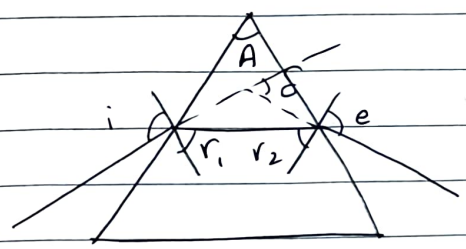
CORRECTION:- bi-focal lenses

top: concave mirror

bottom: convex mirror

6. Prism = 2 triangular bases + 3 rectangular lateral surfaces

∠ b/w 2 lateral surface = Angle of Prism



∠δ = Angle of Deviation

= The angle that emergent ray makes w/ the direction of incident ray

7. DISPERSION

↳ Spectrum: Band of coloured components of a light beam

red light : least
violet : most] bends

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Q. How did Newton conclude that white light is made up of 7 colours?

White light:- Any light that has spectrum similar to sunlight = white light

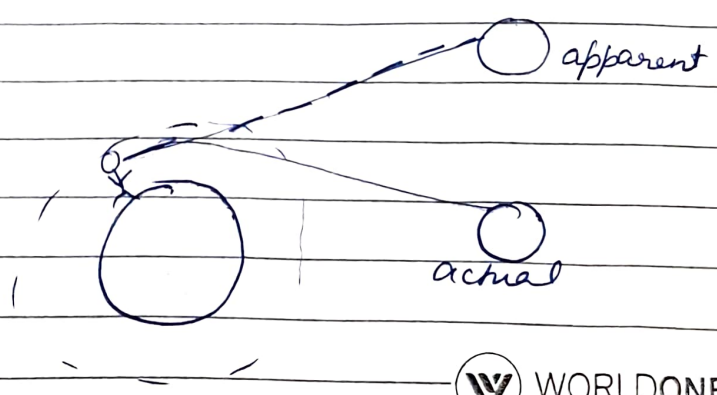
ATMOSPHERIC REFRACTION

1. Twinkling of Stars
↳ Atmospheric cond^n ≠ static
↳ when light from stars reaches our eyes, it bends towards N
↳ ∴ appears higher than what it actually is
↳ Also, amnt. of light entering our eyes changes, sometimes ↑, sometimes ↓ ⟹ twinkling effect

2. Planets:- don't twinkle bcuz, can be considered as extended source of light.
Total deviation averages out to zero by the individual sources of light.

3. Advance sunrise & delayed sunset + app. flattening of sun's disc

CAUSE
Refraction bends towards N



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↳ Scattering

1. Tyndall effect :- When light passes through particles like smoke, dust it gets scattered and its path becomes visible.

2. Sky colour - blue